

TITRATION PRE-LAB QUESTIONS

The following needs to be done before starting the lab in your lab notebooks

- View posted video and as needed refer to your text, as well as internet resources for titration technique.
- Read over Lab (Handout)
- Answer pre-lab questions (primarily using video & also text)

1. Define titration, titrant, equivalence point, end point.
2. The sample is the chemical of unknown concentration. What chemical is the sample in this lab?
3. In this lab, what chemical is the titrant?
4. What chemical goes into the burette?
5. What chemical goes into the flask?
6. When is the **tap** of the burette fully open?
7. What must you ensure, with respect to filling the burette, before beginning the titration?
8. What must you add to your flask before beginning the titration?
9. What colour will the endpoint be, when reached? (be specific)
10. What is a meniscus? Draw a diagram.
11. Where do you read the meniscus? Indicate on your diagram above.
12. To how many decimal places should you read your burette?
13. Why do you place a white sheet under the flask?
14. Show the calculations and describe how you would prepare 250 mL of a 0.100 mol/L solution of NaOH:
 - By diluting a 1.00 M solution
 - By massing out solid NaOH

SCH 3U Lab: Titration

PURPOSE: To find the concentration of vinegar by titration. (We will neutralize a base of known concentration and use math to determine concentration of vinegar)

NOTE: The vinegar will be diluted by a factor of 10 before you receive it.

MATERIALS:

- _____ mol/L Sodium Hydroxide (NaOH) (use concentration calculated above)
- Vinegar of unknown concentration
- 50 mL burette
- 250 mL Erlenmeyer flask
- Phenolphthalein indicator

PROCEDURE:

1. Obtain approximately 60 mL of vinegar in a beaker.
2. Measure out 25.0 mL of the acid using a 25 mL volumetric pipette.
3. Add 1-2 drops of phenolphthalein to the Erlenmeyer flask and position it under the buret.
4. **RECORD THE INITIAL BURET READING TO THE NEAREST 0.1 mL.**
5. Open the buret and begin to add the standard NaOH_(aq) solution slowly. Continually swirl the flask with one hand while controlling the flow of solution with the other.
6. When you notice a pink colour forming in the solution, slow the addition of the standard NaOH_(aq) solution to dropwise.
7. The endpoint is reached when a single drop causes a faint pink colour to remain for a period of 5-10 s.
8. **RECORD THE FINAL BURET READING TO THE NEAREST 0.1 mL.**
9. Repeat the titration for three consistent sets of trial data.

OBSERVATION Table: The Titration of 25.0 mL of diluted vinegar with 0.10 mol/L NaOH

Trial	1	2	3	4
final buret reading (mL)				
initial buret reading (mL)				
volume of NaOH added (mL) = final - initial				
Volume of vinegar	25.0 mL	25.0 mL	25.0 mL	25.0 mL
Colour				

DISCUSSION QUESTIONS

1. How do you know when the acid has been neutralized?
2. Calculate the average volume of base needed to neutralize the acid.
3. Write the balanced equation for the reaction of vinegar and NaOH.
4. Calculate the concentration of the vinegar. SHOW ALL WORK

CONCLUSION:

We determined the concentration of vinegar to be...

What were some challenges you faced in this lab?

Why is it important to run 3 trials in a titration lab?

List two potential sources of error in this lab.